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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student : THIRUKUMARAN S

Programme: **B.E / B.Tech** Slot : D

Course Code: **CSA15** Faculty : Dr.Arul Raj A.M

Course Name: **Cloud Computing and Big Data Analytics** **Date:**

Duration: **1:30 Hr** Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
Total				50	

Student Signature	Faculty Signature	Program Director
Thiru Kumaran S		

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

01. Laas, Paas, & Saas.

Developments in Computing:

* Powerful processors and large memory.
* High-Speed internet and virtualization.
* cloud computing enables flexible resource usage.

Service models:

* laas:

Provides virtual machines, storage and networking, used to move the existing application while controlling OS and Software.

* Paas:

Provides development tools and runtime environments. used to develop new applications without managing Servers.

* Saas:

provides ready-made Software through a browser.
eg: CRM application.

Xaas:

Everything as a Service, including Storage, network, Security and database Services.

Conclusion:

Cloud Services reduce cost and simplify IT management.

02) Virtualization calculation:

Gn:

* Physical server = 15
* CPU utilization = 25%
* old Power = 350w
* ~~old Power~~
* New host power = 450w
* Target utilization = 65%

Step 1: Total CPU load:

15 x 25 = 375%

Step 2:

Active host

375/65 = 5.77 ≈ 6

including N+1 = 7 host

Step: 03 Consolidation ratio:

15/6 = 2.5 : 1

Step 4: utilization:

* All 7 host active = 375/7 = 53.57%
* After one host fail = 375/6 = 62.5%

Step 05: Annual electricity saving:

Old Power = 15 x 350 = 5250 W

new power = 7 x 450 = 3150w

saving = 2.1 x 8760 x 9

₹ 1,65,564

Answer:

6 active host, 7 including N+1, consolidation ratio 2.5:1

Assumption: All host consume rated power continuously and the old server and fully switch off after consolidation.

03. Virtualization Host Calculation:

Given:

* Physical CPU = 24 Cores
* RAM = 96GB
* Hypervisor reserve = 8GB
* VMs = 12
* Each VM = 3 vCPU and 8GB RAM

(a) over-Commitment ratio

Total vCPU = $12 \times 3 = 36$

CPU = $36 / 24 = 1.5 : 1$

Available RAM = $96 - 8 = 88$GB

Configured RAM = $12 \times 8 = 96$GB

RAM = $96 / 88 = 1.09 : 1$

(b) Memory Sufficiency:

Normal = $12 \times 5 = 60$GB → Sufficient.

Peak = $8 \times 8 + 4 \times 6 = 88$GB → Sufficient.

(c) CPU Sufficiency:

Normal = $36 \times 35\% = 12.6$ Cores.

Peak = $36 \times 75\% = 27$ Cores.

Peak CPU demand exceeds 24 cores by 3 cores.

(d) Issues and remedies:

* CPU Contention and slow Performance.
* Memory shortage may cause swapping.
* Security risks due to VM attacks.
* Use load balancing, resource monitoring, access control and VM isolation.

conclusion:

Memory is sufficient, but Peak CPU demand is insufficient.

Q4 Universities - cloud Federation:

=> Cloud federation connects who university data centres

=> conventional cloud architecture provides server, storage and network

=> SDDC enables automatic resources allocation and management.

=> SOA allows sharing of research and digital-library services through APIs

=> Use authentication. authorization, encryption and access control.

=> Administrative safeguards include security policies and regular audits.

=> Resources Allocation. Automatically distributes computing resources between universites.

=> Data backup: Regular backups prevent data loss during failures.

=> Load Balancing: Distributes workloads equally between data centres.

Conclusion: Cloud federation improves healthcare collaboration while protecting Patient information

5. Edge filtering and cloud storage calculation:

Given:

* Cameras = 150
* clip size = 4MB
* one clip every 3 min
* filtering = 25%
* Storage cost = ₹1.50 per GB-month.

(a) without filtering

clips per camera per day:

$(24 \times 60)/3 = 480$

Total clips = $150 \times 480 = ₹2,000$

Data volume:

$₹2,000 \times 4 = 288,000$ MB

= 288GB/day.

Avg bandwith:

$288 \times 8/86400 = 0.02667$ Gbps.

= 26.67 Mbps

(b) storage cost

$288 \times 30 \times 1.50$

= ₹12,960

(c) After edge filtering

uploaded data = 25% of 288 GB

= ₹2 GB/day.

Bandwidth = 6.67 Mbps

Storage Cost:

$₹2 \times 30 \times 1.50$

= ₹ 3,240

Monthly Saving:

= 12960 − 3240

= ₹ 9,720

(d) Serverless Cost:

Monthly invocations:

₹2000 × 30 × 25%. = 540000

cost:

540000 × 0.00015

= ₹81

Rest API:

Transfers video clips between Cameras and cloud storage.

object storage:

Suitable for storing large video files with scalability and durability.